

Practice Project Overview



Estimated Effort: 5 mins

Project Scenario

You have to perform data analytics on a medical insurance charges dataset. This is a filtered and modified version of the [Medical Insurance Price Prediction](#) dataset, available under the [CC0 1.0 Universal License](#) on the [Kaggle](#) website.

Parameters

The parameters used in the dataset are:

1. **Age**

Age of the insured. Integer quantity.

2. **Gender**

Gender of the insured. This parameter has been mapped to numerical values in the following way.

Gender	Assigned Value
Female	1
Male	2

3. **BMI**

Body Mass Index of the insured. Float value quantity.

4. **No_of_Children**

Number of children the insured person has. Integer quantity.

5. **Smoker**

Whether the insured person is a smoker or not. This parameter has been mapped to numerical values in the following way.

Smoker	Assigned Value
Smoker	1
Non smoker	2

6. **Region**

Which region of the USA does the insured belong to. This parameter has been mapped to numerical values in the following way.

Region	Assigned Value
Northwest	1
Northeast	2
Southwest	3
Southeast	4

7. Charges

Charges for the insurance in USD. Floating value quantity.

Objectives

In the next section, you will:

- Load the insurance data as a pandas dataframe
- Clean the data, taking care of the blank entries
- Run exploratory data analysis and identify the attributes that most affect the charges
- Develop single variable and multi variable Linear Regression models for predicting the charges
- Use Ridge regression to refine the performance of Linear regression models.

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